

Unit 2: Phonetics and Phonology

2.1.1 Sound Units & Speech Organs

Before exploring the physiological mechanics of speech production, it is essential to establish the scientific distinction between the two primary sub-fields of sound study in linguistics:

- **Phonetics:** The objective study of human speech sounds from a physical and physiological standpoint. It focuses on how sounds are physically generated by the vocal tract (**articulatory phonetics**), transmitted as sound waves (**acoustic phonetics**), and perceived by the human ear (**auditory phonetics**).
- **Phonology:** The study of the abstract sound systems, structures, and functional patterns within a specific language. While phonetics looks at the raw material (the sounds themselves), phonology looks at how a particular language organizes those sounds to convey meaning.

2.1.1.1 Organs of Speech (Vocal Tract Anatomy)

Human beings do not possess a dedicated biological organ system solely for speech. Every organ used in speech production serves a primary biological survival function, such as respiration, mastication (chewing), or deglutition (swallowing). Speech is an evolutionary, secondary adaptation that utilizes these tracts.

The continuous passage starting from the lungs up to the lips, through which air flows to produce speech sounds, is called the **Vocal Tract**. The speech organs located within this tract are structurally categorized into two functional classes based on their mobility:

1. **Active Articulators:** The flexible, moving organs that alter the shape of the vocal tract or come into contact with other surfaces to modify airflow. The most critical active articulators are the **Tongue** (the most agile organ), the **Lower Lip**, and the **Vocal Folds**.
2. **Passive Articulators:** The stationary, rigid structures forming the upper boundary of the vocal tract. They remain fixed during speech production, acting as target anchors against which active articulators press or approach. Examples include the **Upper Teeth**, the **Alveolar Ridge**, the **Hard Palate**, and the **Velum (Soft Palate)**.

Detailed Anatomical Breakdown of Speech Organs

1. The Respiratory System: Lungs

- **Function:** The lungs serve as the primary power source for speech. They pump an airstream upward through the windpipe (trachea).
- **Linguistic Classification:** Virtually all speech sounds in the English language are generated using a **Pulmonic Egressive Airstream Mechanism**—meaning the airflow is pushed out from the lungs.

2. The Larynx and Vocal Folds

- **Location:** Situated at the top of the trachea, the larynx is a cartilaginous structure popularly known as the "voice box."
- **Mechanism:** Inside the larynx are two horizontal muscular bands known as the **Vocal Folds (or Vocal Cords)**. The space between these folds is called the **Glottis**. The behavior of the glottis determines a foundational phonetic distinction:
 - **Voiced Sounds:** When the vocal folds are brought close together, the rising airstream forces its way through, causing the folds to vibrate rapidly. This vibration produces a buzzing resonance. *Examples: /z/, /v/, /d/, /m/, and all English vowels.*
 - **Voiceless Sounds:** When the vocal folds are pulled wide apart, the glottis remains completely open. The airstream passes through freely without causing any vibration. *Examples: /s/, /f/, /t/, /p/.*

3. The Pharynx

- **Location:** A muscular tube located immediately above the larynx, extending to the back of the oral cavity and nasal cavity. It acts as an initial resonator for the airstream.

4. The Roof of the Mouth (Passive Articulators)

Moving from the front of the upper mouth backward, the roof consists of three distinct zones:

- **Alveolar Ridge (Tooth Ridge):** The firm, bumpy, convex surface located directly behind the upper front teeth. It serves as the passive target for crucial English consonant sounds like /t/, /d/, /n/, /s/, and /l/.
- **Hard Palate:** The smooth, bony, concave structure forming the main roof of the oral cavity. It is crucial for palatal sounds, such as the initial sound in the word *Yes* (/j/).
- **Velum (Soft Palate):** A flexible, muscular tissue hanging directly behind the hard palate. The velum acts as a structural gatekeeper or valve:
 - **Oral Sounds:** When the velum is raised, it seals off the nasal cavity, forcing the entire airstream out through the mouth.
 - **Nasal Sounds:** When the velum is lowered, it opens the nasal passage while the oral passage is blocked, allowing the air to escape through the nose. *Examples:* /m/, /n/, /ŋ/ (as in *sing*).

5. The Tongue (The Primary Active Articulator)

Due to its incredible flexibility, the tongue is divided into distinct operational zones to describe speech production precisely:

- **Tip:** The absolute forward-facing point of the tongue.
- **Blade:** The flat surface immediately behind the tip, sitting right under the alveolar ridge when at rest.
- **Front:** The main body of the tongue located below the hard palate.
- **Back:** The rear portion of the tongue situated opposite the velum.

6. Lips and Teeth

- **Lips:** The lips can be manipulated to block airflow entirely, narrow the opening, or change shape (rounded vs. unrounded). Sounds made using both lips are called **Bilabials** (/p/, /b/, /m/).
- **Teeth:** The upper front teeth interact with the tongue tip to form **Dental** sounds (like the *th* in *thin*, /θ/) or with the lower lip to create **Labiodentals** (/f/, /v/).

2.1.1.2 Phonemes & Allophones

When analyzing speech sounds from a structural perspective, linguists distinguish between the psychological blueprint of a sound and its actual physical pronunciation.

1. The Phoneme (The Psychological Unit)

A **phoneme** is the smallest abstract, structurally distinct unit of sound in a language's mental dictionary that serves to distinguish meaning between words.

Notation Rule: Phonemes are abstract mental categories, and they are always written enclosed within **slanted slashes** / / (e.g., /p/, /t/, /b/).

A phoneme does not possess a singular physical existence; rather, it is a collection of related sounds categorized by the brain as the "same" sound. If you substitute one phoneme for another within a word, the meaning changes completely.

The Minimal Pairs Test

To mathematically prove whether two sounds are independent phonemes in a language, linguists use **Minimal Pairs**. A minimal pair consists of two words that are identical in form except for a single sound segment in the exact same position.

- *Minimal Pair:* **Cat** /kæt/ vs. **Bat** /bæt/
- *Analysis:* Because replacing the initial sound /k/ with /b/ shifts the semantic meaning from a feline animal to a sporting bat, /k/ and /b/ are proven to be distinct **phonemes** in English.
- *Other Examples:* **Pin** /pɪn/ vs. **Bin** /bɪn/, or **Sit** /sɪt/ vs. **Seat** /si:t/.

2. The Allophone (The Physical Variant)

An **allophone** is a concrete, physical variant of a single phoneme. It is the actual sound waves produced by the human vocal tract in physical reality.

Notation Rule: Allophones are objective physical realities, and they are always written enclosed within **square brackets** [] (e.g., [p^h], [p]).

Substituting one allophone for another within a word might sound unusual or give the speaker a foreign accent, but it **never changes the semantic meaning** of the word.

Case Study: The Structural Realizations of English Phoneme /p/

The English phoneme /p/ exists as a single abstract concept in a native speaker's mind, but it is realized physically as three distinct allophones depending on its phonological environment:

1. **Aspirated [p^h]:** When the phoneme /p/ occurs at the absolute beginning of a word (initial position) and is followed by a stressed vowel, it is pronounced with a violent, audible puff of air (aspiration).
 - *Realization:* **Pin** is transcribed phonetically as [p^hɪn].
2. **Unaspirated [p]:** When the same phoneme /p/ occurs immediately following an initial /s/ consonant sound (an s-blend), the vocal tract does not produce a puff of air.
 - *Realization:* **Spin** is transcribed phonetically as [spɪn].
3. **Unreleased [p̚]:** When /p/ occurs at the very end of a word (final position), the lips close to form the sound but do not physically release the air trap.
 - *Realization:* **Cap** is transcribed phonetically as [kæp̚].

Key Structural Differences: Phonemes vs. Allophones

Dimension	Phoneme	Allophone
Level of Reality	Abstract / Psychological (Exists in the mind)	Concrete / Phonetic (Exists in speech waves)
Transcription Brackets	Slanted Slashes / /	Square Brackets []
Impact on Meaning	Contrastive: Changing it alters word meaning.	Non-Contrastive: Changing it does not alter meaning.
Language System	Universal to the language grammar system.	Dependent on the exact position inside a word.
Testing Mechanism	Discovered using the Minimal Pairs Test .	Discovered by analyzing structural environment distribution.

Patterns of Allophonic Distribution

Allophones are distributed within a language according to strict environmental rules, falling into two primary structural patterns:

1. Complementary Distribution

This occurs when allophones are mutually exclusive; where one allophone appears, the other can never appear. They occupy completely separate environmental niches.

- *Analogy:* Consider the role of a store manager and an assistant manager. When the manager is present, they take the lead; when they are away, the assistant takes over. They never occupy the same space simultaneously.
- *Linguistic Example:* The aspirated allophone [p^h] only occurs at the start of a word. The unaspirated [p] never occurs at the start of a word if an /s/ precedes it. Their environments complement each other perfectly.

2. Free Variation

This occurs when two distinct allophones of the same phoneme can be freely interchanged in the exact same environmental position without altering the meaning or sounding incorrect. The choice is left up to the speaker's personal preference or speed.

- *Linguistic Example:* In pronouncing the final sound of the word "Map," a speaker can choose to release the lips cleanly, producing [mæp], or leave the lips closed, producing [mæp̚]. Both variants are perfectly acceptable in free variation.

2.1.2 Speech Sound Classification

In English phonetics, the standard alphabet of 26 letters does not match the actual spoken sound system. The English sound system consists of **44 distinct phonemes**. These 44 speech sounds are broadly divided into two major operational categories based on how the airstream is modified as it passes through the vocal tract:

- **Vowels (20 sounds):** Speech sounds produced with a completely open vocal tract, where there is **no obstruction, narrowing, or blockage** of the airstream along the mid-line of the oral cavity.
- **Consonants (24 sounds):** Speech sounds produced by **obstructing, constricting, or redirecting** the airstream at one or more points within the vocal tract by an active articulator meeting a passive articulator.

2.1.2.1 Classification of English Vowels

The 20 English vowel sounds are sub-classified into two structural groups based on whether the sound remains pure and static or shifts dynamically during pronunciation:

1. **Monophthongs (12 Pure Vowels):** Vowels whose quality remains constant, static, and unchanged throughout their entire production duration (e.g., the sound in *see* or *cat*).
2. **Diphthongs (8 Gliding Vowels):** Vowels that begin with one vocalic quality and smoothly glide or shift into another vocalic quality within the boundaries of a single syllable (e.g., the sound in *buy* or *coin*).

Criteria for Classifying Monophthongs (Pure Vowels)

Linguists classify the 12 pure vowels using a three-dimensional grid based on the physical position and shape of the tongue and lips:

1. Tongue Height (Vertical Axis)

This measures how close the highest point of the tongue is pulled up toward the roof of the mouth:

- **Close (High) Vowels:** The tongue is elevated close to the palate. *Examples:* /i:/ (as in *see*), /u:/ (as in *too*).
- **Mid Vowels (Half-Close / Half-Open):** The tongue sits in an intermediate position. *Examples:* /e/ (as in *bed*), /ə/ (the neutral schwa sound in *about*).
- **Open (Low) Vowels:** The tongue is lowered in the mouth, and the jaw is dropped wide open. *Examples:* /æ/ (as in *cat*), /ɑ:/ (as in *father*).

2. Tongue Position / Part of the Tongue (Horizontal Axis)

This measures which specific horizontal region of the tongue is raised to its highest point:

- **Front Vowels:** The front body of the tongue is raised toward the hard palate. *Examples:* /i:/, /ɪ/, /e/, /æ/.
- **Central Vowels:** The middle or central body of the tongue is raised. *Examples:* /ə/, /ɜ:/ (as in *bird*), /ʌ/ (as in *cup*).
- **Back Vowels:** The back of the tongue is pulled backward and upward toward the velum. *Examples:* /u:/, /ʊ/ (as in *book*), /ɔ:/ (as in *door*), /ɒ/ (as in *hot*), /ɑ:/.

3. Lip Position (Rounding Axis)

This tracks the physical shape of the lips during articulation:

- **Rounded Vowels:** The lips form a circular, pushed-forward shape. *Examples:* Back vowels like /u:/, /ʊ/, /ɔ:/, /ɒ/.
- **Unrounded (Spread/Neutral) Vowels:** The lips are either flattened, pulled out to the sides, or completely relaxed. *Examples:* Front and central vowels like /i:/, /ɪ/, /e/, /æ/, /ə/.

4. Vowel Length (Tenseness)

- **Long Vowels (5 sounds):** Articulated with greater muscular tension and duration. Written with a colon mark :. *Examples:* /i:/, /ɑ:/, /ɔ:/, /u:/, /ɜ:/.
- **Short Vowels (7 sounds):** Articulated quickly with minimal muscular tension. *Examples:* /ɪ/, /e/, /æ/, /ɒ/, /ʊ/, /ʌ/, /ə/.

The 8 English Diphthongs (Gliding Vowels)

Diphthongs are classified based on the direction of their structural movement toward the final target vowel slot:

- **Centring Diphthongs (Glide toward the central neutral Schwa sound /ə/):**
 - /ɪə/ — As heard in *Clear, Fear, Near*
 - /ʊə/ — As heard in *Cure, Pure, Tour*
 - /eə/ — As heard in *Hair, Care, Stair*
- **Closing Diphthongs (Glide upward toward a close vowel):**
 - *Gliding toward Close Front /ɪ/:*
 - /eɪ/ — As heard in *Face, Day, Rain*
 - /aɪ/ — As heard in *Price, Fly, High*
 - /ɔɪ/ — As heard in *Boy, Coin, Voice*
 - *Gliding toward Close Back /ʊ/:*
 - /əʊ/ — As heard in *Go, Home, Boat*
 - /aʊ/ — As heard in *Mouth, Cow, Loud*

2.1.2.2 Classification of English Consonants

The 24 English consonant sounds cannot be classified by tongue height. Instead, they are mapped systematically using a rigorous **Three-Term Label** system that accounts for three specific phonetic parameters:

Consonant Description = Voicing + Place of Articulation + Manner of Articulation

Parameter 1: Voicing State (The Glottis Condition)

- **Voiceless (9 sounds):** The vocal folds do not vibrate.
- **Voiced (15 sounds):** The vocal folds vibrate rapidly, adding vocal weight.

Parameter 2: Place of Articulation (Where the sound is made)

This designates the exact geographical spot in the vocal tract where the active articulator makes contact with or approaches a passive articulator:

1. **Bilabial:** Formed by bringing both lips together. *Sounds:* /p/, /b/, /m/, /w/.
2. **Labiodental:** Formed by bringing the lower lip into light contact with the upper front teeth. *Sounds:* /f/, /v/.
3. **Dental:** Formed by placing the tip or blade of the tongue against the inner surface of the upper front teeth. *Sounds:* /θ/ (voiceless *th* in *thin*), /ð/ (voiced *th* in *this*).
4. **Alveolar:** Formed by placing the tongue tip or blade against the rough alveolar tooth ridge. *Sounds:* /t/, /d/, /s/, /z/, /n/, /l/.
5. **Post-Alveolar:** Formed by placing the tongue tip just behind the alveolar ridge, near the edge of the hard palate. *Sound:* /r/.

6. **Palato-Alveolar (Alveolo-Palatal):** Formed by the tongue blade approaching the alveolar ridge while the main front body of the tongue rises toward the hard palate. *Sounds: /ʃ/ (sh in shoe), /ʒ/ (s in measure), /tʃ/ (ch in church), /dʒ/ (j in judge).*
7. **Palatal:** Formed by bringing the front of the tongue close to the hard bony palate. *Sound: /j/ (the y sound in yes).*
8. **Velar:** Formed by raising the back of the tongue against the soft palate (velum). *Sounds: /k/, /g/, /ŋ/ (the ng sound in sing).*
9. **Glottal:** Formed entirely within the throat, using the space between the open vocal folds (glottis) to create internal friction. *Sound: /h/.*

Parameter 3: Manner of Articulation (How the sound is made)

This describes the structural path the air takes and how much the passage is squeezed or blocked:

1. **Plosives (Stops):** The air is completely blocked inside the mouth, air pressure builds up behind the closure, and then the articulators separate suddenly, releasing the air with a tiny burst or pop. *Sounds: /p/, /b/, /t/, /d/, /k/, /g/.*
2. **Fricatives:** The articulators are brought very close together, leaving only an extremely narrow escape channel. As the air is forced out, it produces constant, turbulent friction. *Sounds: /f/, /v/, /θ/, /ð/, /s/, /z/, /ʃ/, /ʒ/, /h/.*
3. **Affricates:** A structural combination of a plosive and a fricative. The sound begins with a total air blockage (plosive stage) but releases slowly into a narrow friction channel (fricative stage). *Sounds: /tʃ/, /dʒ/.*
4. **Nasals:** The oral cavity is completely blocked at some point, but the velum (soft palate) is lowered, allowing the entire air supply to escape freely through the nasal cavity. *Sounds: /m/, /n/, /ŋ/.*
5. **Lateral:** The center of the tongue blocks the middle path of the mouth (by touching the alveolar ridge), forcing the air to flow out sideways over the dropped lateral edges of the tongue. *Sound: /l/.*
6. **Approximants (Glides / Semi-vowels):** The articulators approach each other but remain far enough apart that air escapes without producing turbulent friction. They act structurally like consonants but are articulated like vowels. *Sounds: /w/, /r/, /j/.*

Complete 24 English Consonants Mapping Matrix

Use this precise reference chart to quickly identify the full structural profile of any English consonant:

Place →Manner ↓	Bilabial	Labiodental	Dental	Alveolar	Palato-Al veolar	Palatal	Velar	Glottal
Plosive	/p/, /b/			/t/, /d/			/k/, /g/	
Fricative		/f/, /v/	/θ/, /ð/	/s/, /z/	/ʃ/, /ʒ/			/h/
Affricate					/tʃ/, /dʒ/			
Nasal	/m/			/n/			/ŋ/	
Lateral				/l/				
Approximant	/w/				/r/	/j/		

Note: Within the cell pairs, the symbol on the left represents a **Voiceless** sound, and the symbol on the right represents a **Voiced** sound.

2.1.3 Prosody & Syllable Structure

While individual sound segments (vowels and consonants) are known as **segmental phonemes**, speech also involves features that span across larger stretches of utterances—such as syllables, words, and entire sentences. These features are called **suprasegmental features** or **Prosody**. Prosody includes the structural organization of speech chunks, rhythm, musical pitch, and voice loudness.

2.1.3.1 Syllables (Structure & Division)

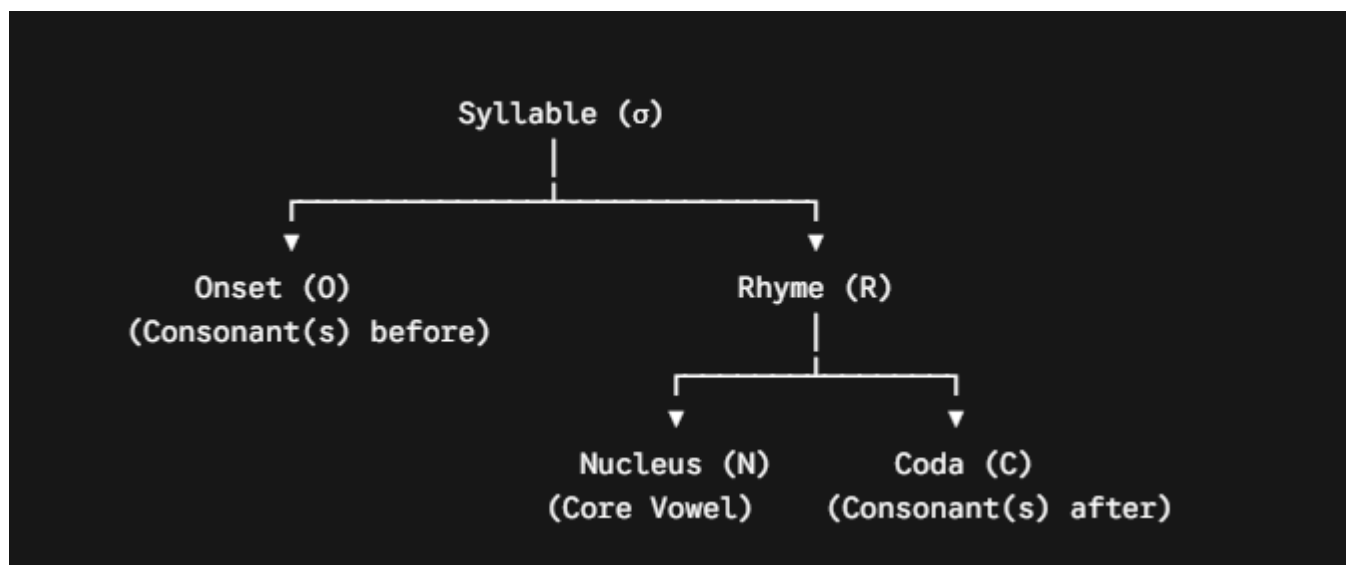
1. What is a Syllable?

A **syllable** is a fundamental unit of speech organization that is larger than a single sound segment but typically smaller than a whole word. From a physical perspective, it corresponds to a single, uninterrupted push of air from the lungs, producing a peak of sonority (loudness).

Every syllable must contain a **vocalic element** (a vowel or a vowel-like sound) at its core. You cannot have a syllable without a vowel, but you *can* have a syllable without a consonant (e.g., the word *Eye* /aɪ/).

2. The Internal Anatomy of a Syllable

Linguists analyze the internal structure of a syllable hierarchically. A syllable (represented by the Greek letter Sigma \sigma) is split into two primary components: the **Onset** and the **Rhyme**. The Rhyme is further split into the **Nucleus** and the **Coda**.



- **Onset (O):** The initial consonant or consonants occurring at the very beginning of the syllable, before the core vowel. (Optional in English structure).
- **Rhyme (R):** The core framework of the syllable, grouping the vowel and any following consonants together.
- **Nucleus (N):** The absolute essential core of the syllable, typically occupied by a **vowel** (monophthong or diphthong). It carries the highest loudness and pitch.
 - *Note on Syllabic Consonants:* Occasionally, highly sonorous consonants like /l/, /n/, or /m/ can act as the nucleus in unstressed final syllables, such as the final slot in *Button* ['bʌt.n̩] or *Bottle* ['bɒt.l̩].
- **Coda (C):** The consonant or consonants that follow the nucleus, closing out the syllable. (Optional in English structure).

Example Structural Mappings:

- **Cat** /kæt/ → Onset = /k/, Nucleus = /æ/, Coda = /t/
- **It** /ɪt/ → Onset = Zero (Empty), Nucleus = /ɪ/, Coda = /t/
- **See** /si:/ → Onset = /s/, Nucleus = /i:/, Coda = Zero (Empty)

3. Consonant Clusters & Constraints

English allows multiple consonants to group together within a single onset or coda. These groupings are called **Consonant Clusters**.

- **Max Onset Limit:** English allows up to three consonants in a single onset, but only follow strict structural laws (Phonotactics). The first must be /s/, the second a voiceless plosive (/p/, /t/, /k/), and the third a liquid/glide (/l/, /r/, /j/, /w/). Example: *Splash* /splæʃ/, *Street* /stri:t/.
- **Max Coda Limit:** English can allow up to four consonants in a single coda, usually occurring when grammatical endings are attached. Example: *Twelfths* /twelfθs/.

4. Syllable Division (Syllabification)

When dividing words with multiple syllables, linguists use the **Maximal Onset Principle (MOP)**. This rule states that when a consonant sits between two vowels, it should be assigned to the onset of the second syllable rather than the coda of the first, as long as it forms a valid consonant combination allowed by English grammar rules.

- *Example:* The word "**Extremist**" is syllabified as /ek.'stri:mɪst/ (not /eks.tri:m.ɪst/).

2.1.3.2 Stress (Word & Sentence stress)

Stress is the relative degree of force, muscular energy, and loudness applied to a specific syllable during speech. A stressed syllable is physically characterized by **higher pitch, longer duration, and clearer vowel quality**.

1. Word Stress

In English, word stress is **distinctive** (phonemic); changing the location of the stress can alter the grammatical class or meaning of the word.

The Noun-Verb Paradigm:

Many two-syllable words in English shift their identity entirely based on whether the stress falls on the first or second syllable: →

- **'In-crease** (Stress on 1st syllable) → **Noun** (e.g., "There is an 'increase in production.")
- **In-'crease** (Stress on 2nd syllable) → **Verb** (e.g., "We need to in-'crease production.")
- *Other Examples:* **'Ob-ject** (Noun) vs. **Ob-'ject** (Verb); **'Pre-sent** (Noun) vs. **Pre-'sent** (Verb).

2. Sentence Stress

Sentence stress shifts the focus away from individual vocabulary rules toward communication context. It determines which word carries the core weight of the message within an utterance. English is a **stress-timed language**, meaning stressed syllables occur at relatively regular time intervals, forcing unstressed syllables to crush together.

- **Content Words:** Nouns, main verbs, adjectives, and adverbs normally receive stress because they carry the primary information.
- **Function Words:** Articles, prepositions, pronouns, and auxiliary verbs are normally unstressed and reduced.

Contrastive Stress:

Sentence stress can be shifted dynamically to highlight a specific contrast or correct a misunderstanding:

- "'**John** bought a new car." (It was John, not someone else).
- "John '**bought** a new car." (He purchased it; he didn't rent it).
- "John bought a '**new** car." (The car is brand new, not used).

2.1.3.3 Intonation (Pitch & Tone)

Intonation refers to the meaningful variation of the pitch of the voice across an entire sentence or utterance. If word stress provides the rhythmic heartbeat of English, intonation provides its melody.

The functional unit of intonation is the **Tone Unit**. Within every tone unit, there is one highly prominent syllable where the pitch changes direction dramatically; this is called the **Tonic Syllable** (or Nucleus).

English relies on three primary intonation directions:

1. Falling Tone (↘)

The pitch of the voice drops down cleanly on the tonic syllable. This tone signals **finality, completeness, certainty, and authority**.

- **Declarative Statements:** "*We are leaving today.*" ↘
- **Wh- Questions:** "*Where do you live?*" ↘
- **Commands:** "*Sit down right now.*" ↘

2. Rising Tone (↗)

The pitch of the voice glides upward on the tonic syllable. This tone signals **incompleteness, uncertainty, politeness, or an invitation for validation**.

- **Yes/No Questions:** "*Are you coming tomorrow?*" ↗
- **Polite Requests:** "*Could you pass the salt?*" ↗
- **Lists (Incomplete items):** "*I need milk ↗, eggs ↗, and bread ↘.*"

3. Fall-Rise Tone (↘↗)

The pitch drops down initially and then curves back upward within a single syllable or word chunk. This tone signals **reservation, hesitation, doubt, hidden meanings, or politeness**.

- **Example:** "*Is she a good doctor? → "Well, she's **clever**..." ↘↗* (Implies: She is smart, but I have doubts about her actual clinical skills or behavior).

Core Prosody Exam Summary Reference

Component	Level	Core Acoustic Metric	Primary Linguistic Function
Syllable	Structural Chunk	Sonority / Vowel Core	Organizes phonemes into pronounceable physical units.
Word Stress	Lexical Level	Loudness / Muscular Force	Differentiates grammatical categories (Noun vs. Verb).
Sentence Stress	Utterance Level	Rhythmic Prominence	Identifies focus points and contrastive information.
Intonation	Sentence Level	Pitch Contour (Hz)	Expresses speaker attitude, emotion, and sentence type.